

IMO Training 2020
Inversion
3rd October, 2020

Source: *107 Geometry Problems* by Titu Andreescu. The series of *106 Geometry Problems* and this book is the same source as most of the notes in level 1 by KKK.

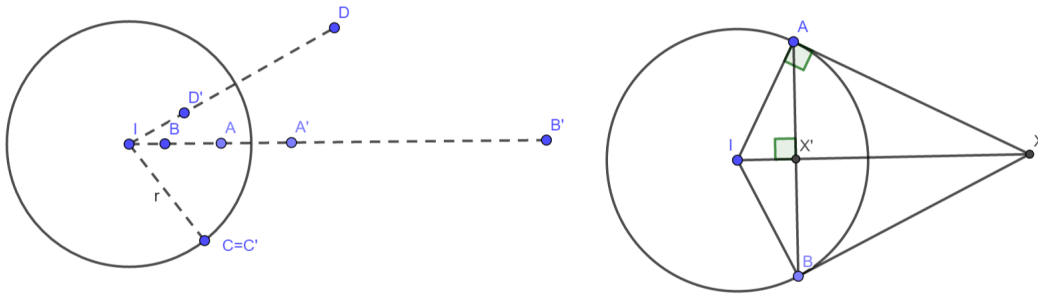
A free source online would be the book *Euclidean Geometry in Mathematical Olympiads* by Evan Chen, where the chapter on inversion is legally free (https://www.maa.org/sites/default/files/pdf/ebooks/pdf/EGMO_chapter8.pdf).

Of course, another source would be the *Projective Geometry and Inversion* in our IMO training google drive under Additional Notes→有生之年系列→Intermediate Notes.

1 Inversion

Given a circle ω with centre I and radius $r > 0$, we define the image X' of point X under inversion about ω as the point on ray IX such that

$$IX \cdot IX' = r^2.$$



Notice that

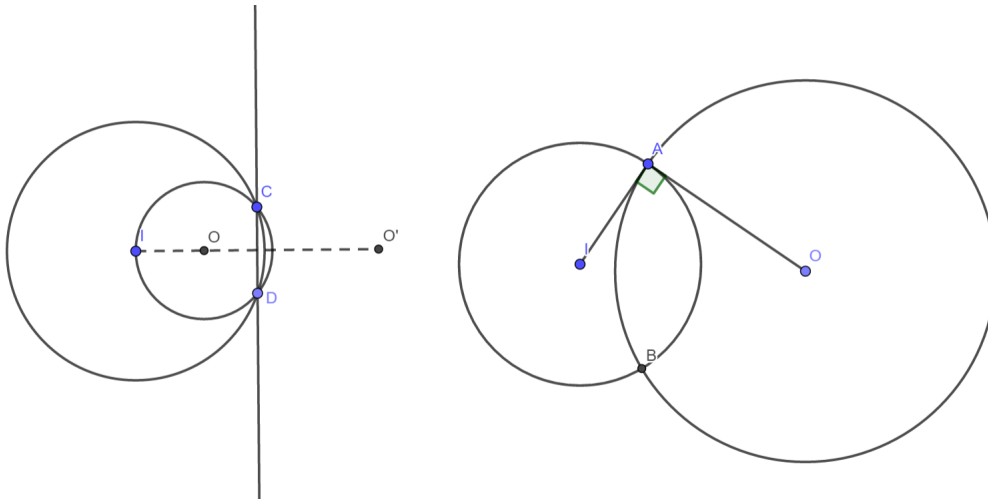
1. The points inside ω (i.e. $IX < r$) are mapped to the outside (i.e. $IX > r$); vice versa.
2. Inversion about the same circle twice is an identity mapping, i.e. $X'' = X$. In other words, X' is the image of X if and only if X is the image of X' .
3. Points lying on ω are exactly those being invariant under inversion, i.e. $X' = X$.
4. Conventionally, the image of I is ∞ and the image of ∞ is I .

The ruler-compass construction is as the right figure above, which follows from $\triangle IAX' \sim \triangle IXA$. In geogebra, it is the "reflection about circle" action.

As we will see, the radius of inversion may often be chosen arbitrarily, therefore there would often be notions of "inverting about a point", meaning inverting about a circle with that point as centre and an arbitrary radius.

1.1 Properties

1. Let I, X, Y be pairwise distinct non-collinear points. Denote by X' and Y' the images of X and Y under inversion about a circle with centre I and radius $r > 0$.
 - (a) $\triangle IXY \sim \triangle IY'X'$
 - (b) X, Y, Y', X' concyclic
 - (c) (Angle chasing) $\angle XYI = \angle IX'Y'$
 - (d) (Length chasing) $X'Y' = XY \cdot \frac{r^2}{IX \cdot IY}$ and $XY = X'Y' \cdot \frac{r^2}{IX' \cdot IY'}$
2. (Image of line) Denote by ℓ' the image of line ℓ under inversion about I .
 - (a) If ℓ passes through I , then $\ell' = \ell$
 - (b) If ℓ does not pass through I , ℓ' is a circle with center O passing through I such that $OI \perp \ell$.
3. (Image of circle) Denote by ω' the image of line ω with centre O under inversion about a circle Γ with centre I and radius $r > 0$.
 - (a) If ω passes through I , then ω' is a line perpendicular to OI .
 - i. O' is the reflection of I about line ω' .
 - (b) If ω does not pass through I , then ω' is a circle.
 - i. In general, O' is not the centre of ω' . Still, the centre of ω' and the points O, O' all lie on the same ray from I .
 - (c) Suppose ω and Γ intersect at points A and B . Then $\omega' = \omega$ if and only if $\angle OAI = 90^\circ$. In this case, we say ω and Γ are *orthogonal*.



1.2 Equivalent Statements

A basic usage of inversion is to invert the whole question (including the statement to be proved) to obtain an equivalent problem, and sometimes the equivalent problem is easier to solve.

1. (Ptolemy's Inequality) Let A, B, C, D be four points on the plane. Prove that

$$AB \cdot CD + AD \cdot BC \geq AC \cdot BD.$$

Show that equality case holds if and only if A, B, C, D are either concyclic or collinear in this order.

2. (IMO 2003 Shortlist) Let $\Gamma_1, \Gamma_2, \Gamma_3, \Gamma_4$ be distinct circles such that Γ_1, Γ_3 are externally tangent at P , and Γ_2, Γ_4 are externally tangent at the same point P . Suppose that Γ_1 and Γ_2, Γ_2 and Γ_3, Γ_3 and Γ_4, Γ_4 and Γ_1 meet at A, B, C, D , respectively, and that all these points are different from P . Prove that

$$\frac{AB \cdot BC}{AD \cdot DC} = \frac{PB^2}{PD^2}.$$

3. (IMO 1996) Let P be a point inside $\triangle ABC$ such that

$$\angle APB - \angle ACB = \angle APC - \angle ABC.$$

Let D, E be the incentres of $\triangle APB, \triangle APC$ respectively. Show that the lines AP, BD, CE meet at a point.

4. (Romania TST 2008) Let ABC be a triangle with $\angle BAC < \angle ACB$. Let D, E be points on the sides AC and AB , respectively, such that $\angle ACB = \angle BED$. If F lies in the interior of the quadrilateral $BCDE$ such that the circumcircle of $\triangle BCF$ is tangent to the circumcircle of $\triangle DEF$, and the circumcircle of $\triangle BEF$ is tangent to the circumcircle of $\triangle CDF$, prove that the points A, C, E, F are concyclic.
5. (IMO 2018) A convex quadrilateral $ABCD$ satisfies $AB \cdot CD = BC \cdot DA$. Point X lies inside $ABCD$ so that $\angle XAB = \angle XCD$ and $\angle XBC = \angle XDA$. Prove that $\angle BXA + \angle DXC = 180^\circ$.

1.3 Choice of inversion circle

A more sophisticated usage of inversion is to consider some effects of inversion without inverting the whole configuration. In this case, a wise choice of circle is crucial.

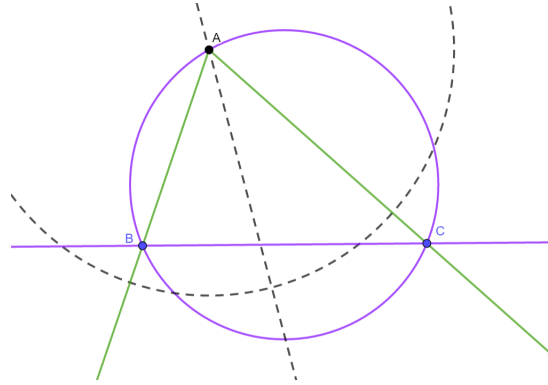
1. (Iran 1995) Let M, N, P be the points where the incircle of scalene triangle ABC touches its sides BC, CA, AB respectively. Prove that the orthocentre of $\triangle MNP$, the incentre I of $\triangle ABC$ and the circumcentre O of $\triangle ABC$ are collinear.
2. (Feuerbach's Theorem) Show that the nine-point circle of a triangle is tangent to its incircle and three excircles.

2 Crazy inversion

Given a $\triangle ABC$, we consider the transformation which first reflects point X over the A -angle bisector, then inverts about A with radius $\sqrt{AB \cdot AC}$ into X' . We call X' the image of X in $\sqrt{bc}$ -inversion. Complicated at the first glance, but such a trick is "tailor-made" to the triangle such that

1. B maps to C and C maps to B ; and
2. the circumcircle of $\triangle ABC$ maps to line BC , line BC maps to the circumcircle of $\triangle ABC$.

Note that lines AX and AX' are isogonal, so it is comfy to use such a trick in isogonal lines.



2.1 Isogonality

1. (Dumpty-point) In $\triangle ABC$, let S be a point on the circumcircle of $\triangle ABC$ such that AS is the A -symmedian. Prove that the midpoint of AS is exactly the A -dumpty point, i.e. the point Q satisfying $\angle BAQ = \angle ACQ$ and $\angle CAQ = \angle ABQ$.
2. (Mixtilinear Incircle) Let ω be the circumcircle of $\triangle ABC$. Circle ω_1 is tangent to AB and AC , and it touches ω internally at T . Let D be the point of tangency of BC and the A -excircle. Show that $\angle BAT = \angle DAC$.
3. (Serbia 2008) Given a $\triangle ABC$. Points D, E lie on line AB such that $AD = AC$, $BE = BC$, and the points D, A, B, E are collinear in this order. Bisectors of internal angles at A and B intersect BC, AC at P and Q , and the circumcircle of $\triangle ABC$ at M and N respectively. Line through A and the centre O_1 of the circumcircle of $\triangle BME$ and line through B and the centre O_2 of the circumcircle of $\triangle AND$ intersect at X . Prove that $CX \perp PQ$.

2.2 Create your own inversion

1. (IMO 2015) Let ABC be an acute triangle with $AB > AC$. Let Γ be its circumcircle, H its orthocentre, and F the foot of the altitude from A . Let M be the midpoint of BC . Let Q be the point on Γ such that $\angle HQA = 90^\circ$ and let K be the point on Γ such that $\angle HKQ = 90^\circ$. Assume that the points A, B, C, K, Q are all different and lie on Γ in this order. Prove that the circumcircles of $\triangle KQH$ and $\triangle FKM$ are tangent to each other.